# IMPLEMENTATION OF HALF-ADDER AND FULL-ADDER

## OBJECTIVE

To verify the functionality of Half-Adder and Full-Adder circuits through the truth table and output validation.

## THEORY

### Adders:

- The most basic arithmetic operation, no doubt is addition of two binary digits.

### Half Adder:

- A combinational circuit that performs addition of two bits is called Half Adder. This circuit needs two binary inputs and two outputs. The output variable produces sum and carry.

### Truth table:

| X | Y | C (carry) | S (sum) |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

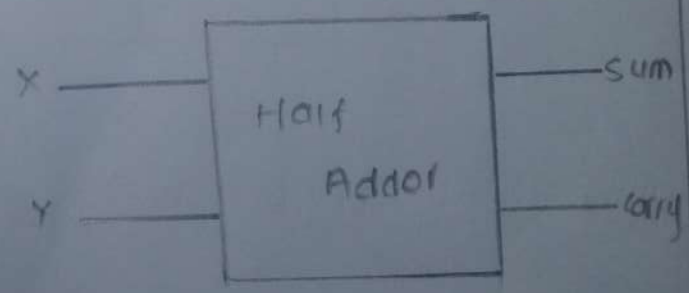



fig: Block diagram of half adder.

## Full-Adder

- A combinational circuit that performs the addition of three bits (two significant bits and a previous carry) is called a full adder. Two significant bit represented by A and B (two of input variables) to be added, C represents carry from previous lower significant position.

### Truth table

| Decimal No | A | B | C | c (carry) | s (sum) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 1 |
| 2 | 0 | 1 | 0 | 0 | 1 |
| 3 | 0 | 1 | 1 | 1 | 0 |
| 4 | 1 | 0 | 0 | 0 | 1 |
| 5 | 1 | 0 | 1 | 1 | 0 |
| 6 | 1 | 1 | 0 | 1 | 0 |
| 7 | 1 | 1 | 1 | 1 | 1 |

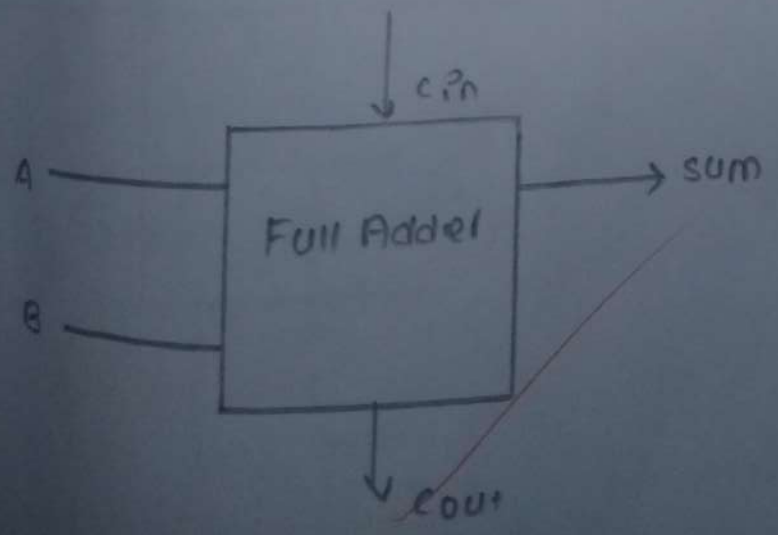



fig. Block diagram of full Adder.

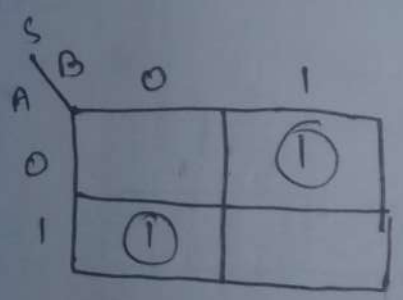
## OBSERVATION AND DIAGRAMS

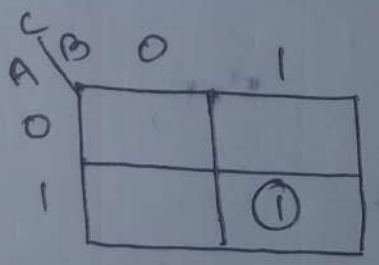
### HALF-Adder

Truth table

| Input | | Expected output | | Observed output | |
|---|---|---|---|---|---|
| A | B | $S_e$ | $C_e$ | $S_o$ | $C_o$ |
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 | 0 | 1 |

K-MAP




$S = \bar{A}B + A\bar{B}$

$= A \oplus B$

$C = AB$

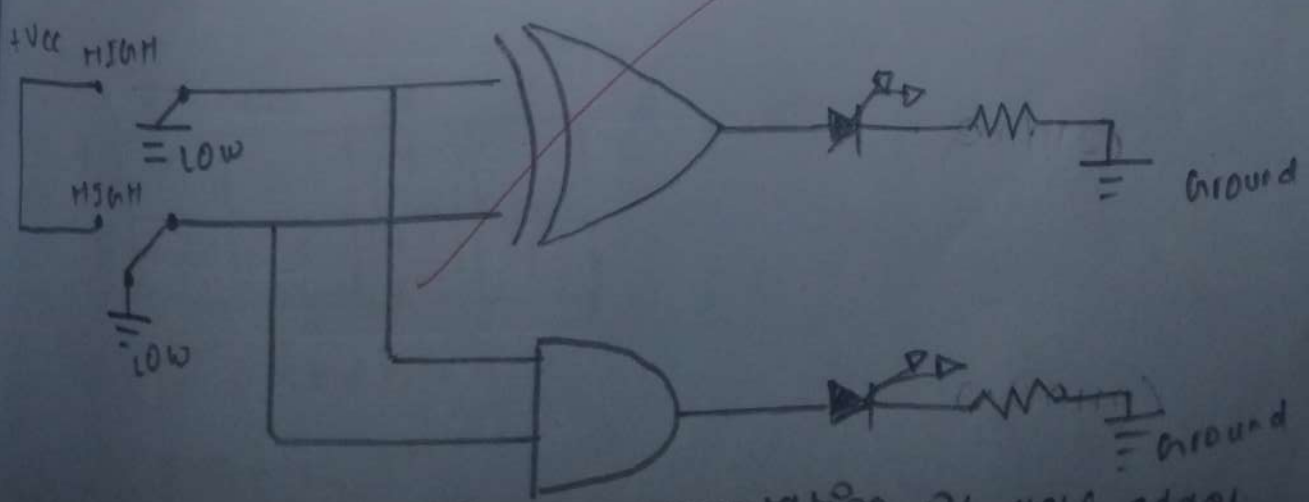



fig: Implementation of Half adder

Full Adder

| Input | | | Expected output | | Observed output | |
|---|---|---|---|---|---|---|
| A | B | C | $S_e$ | $C_e$ | $S_o$ | $C_o$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |

k-map

S

| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | 1 | | 1 |
| 1 | 1 | | 1 | |

$S = A'B'C + A'BC' + AB'C' + ABC$

$= (A \oplus B) \oplus C$

C

| 0 | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | | 1 | |
| 1 | | 1 | 1 | 1 |

$C = BC + AC + AB$

$= AB + BC + AC$

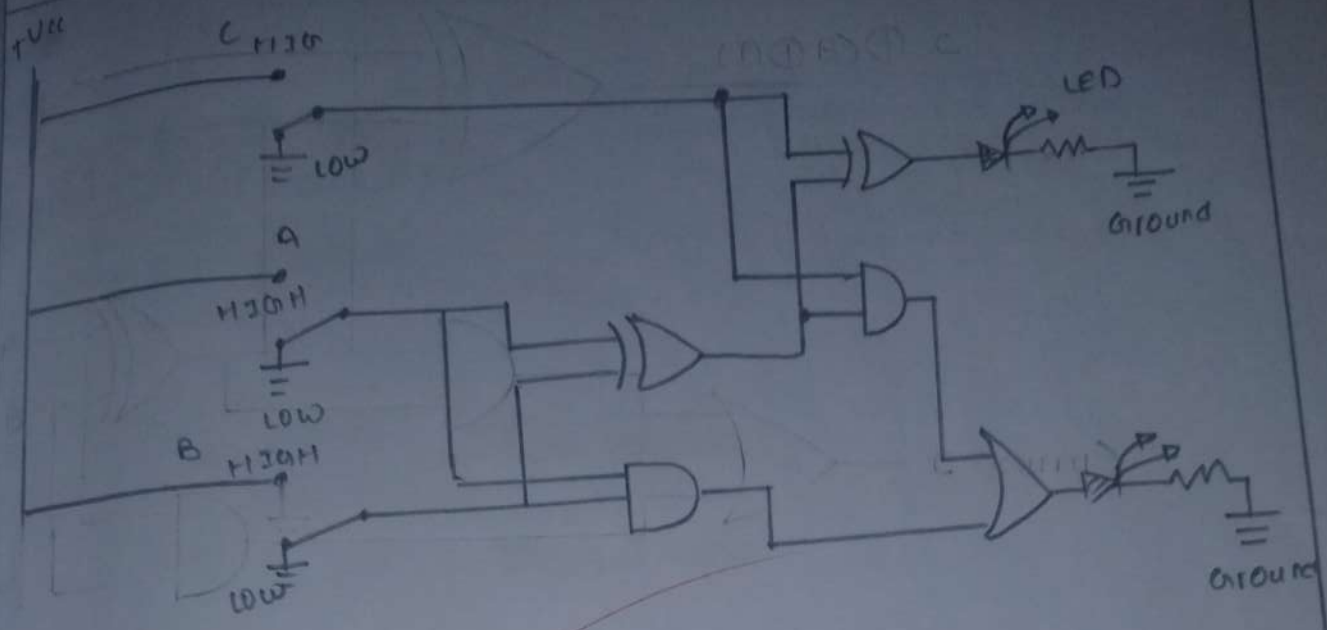



fig: Implementation of full-Adder.

## Discussion and conclusion

In this lab, we learnt about Adders (i.e Half Adder and Full Adder). we observed the sum and carry in both adders and learnt How Adders works and how to represent Adders in diagram.